OCR A-Level Computer Science

H446/01 Computer Systems

Practice Paper

Centre Number	
Candidate Number	
Candidate Name	

INSTRUCTIONS TO CANDIDATES

- Time allowed: 1 hour 15 minutes
- Total marks for this paper: 40
- Write your answers in the spaces provided
- Answer ALL questions

INFORMATION FOR CANDIDATES

- The marks for each question are shown in brackets [ ]
- The quality of your written communication will be assessed in your answers
- This document consists of 8 pages

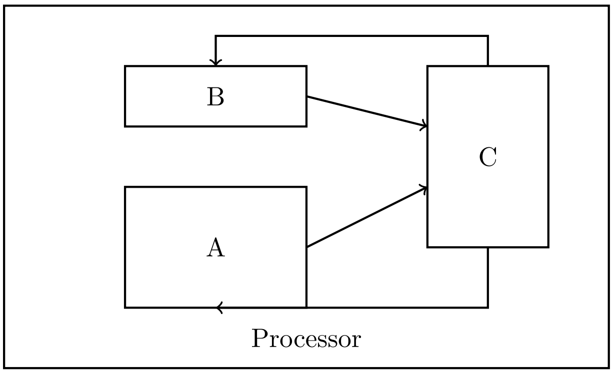
Answer ALL questions

1. (a) Define the term 'clock speed' as it relates to processors. [2 marks]

   (b) Explain how increasing the clock speed affects the performance of a processor. [2 marks]

   (c) A processor has a clock speed of 3.2 GHz. Calculate how many clock cycles occur in 0.5 seconds. Show your working. [2 marks]

2. The diagram below shows a simplified structure of a processor.



(a) Identify the components labeled A, B, and C in the diagram. [3 marks]

(b) Describe the function of component A. [2 marks]

(c) Explain how components A, B, and C work together during the fetch-decode-execute cycle. [4 marks]

3. (a) Describe what is meant by a 'multi-core processor'. [2 marks]

   (b) Explain two advantages of having multiple cores in a processor. [4 marks]

   (c) Explain one reason why doubling the number of cores does not always double the performance of a computer system. [2 marks]

4. (a) Define the term 'cache memory'. [2 marks]

   (b) Modern processors typically have different levels of cache (L1, L2, L3).

      (i) State which level of cache is typically the fastest. [1 mark]

      (ii) Explain why processors have different levels of cache rather than just one large cache. [3 marks]

   (c) Describe how the principle of locality relates to cache memory performance. [3 marks]

5. Compare and contrast CISC (Complex Instruction Set Computer) and RISC (Reduced Instruction Set Computer) processor architectures. In your answer, you should include:

   - The key characteristics of each architecture
   - Advantages and disadvantages of each
   - Examples of where each might be used

   [8 marks]

Grade Boundaries

Grade	A*	A	B	C	D
Marks	36-40	32-35	28-31	24-27	20-23